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| Tool Version : Vivado v.2024.1 (win64) Build 5076996 Wed May 22 18:37:14 MDT 2024
| Date   : Wed Oct 21 02:08:38 2026
| Host   : MICHALADENND5D2 running 64-bit major release (build 9200)
| Command   : report_utilization -file lab4_top_utilization_synth.rpt -pb lab4_top_utilization_synth.pb
| Design   : lab4_top
| Device   : xc7a100tcsg324-1
| Speed File   : -1
| Design State : Synthesized

Utilization Design Information

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1. Slice Logic

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs*	248	0	0	63400	0.39
LUT as Logic	248	0	0	63400	0.39
LUT as Memory	0	0	0	19000	0.00
Slice Registers	197	0	0	126800	0.16
Register as Flip Flop	197	0	0	126800	0.16
Register as Latch	0	0	0	126800	0.00
F7 Muxes	0	0	0	31700	0.00
F8 Muxes	0	0	0	15850	0.00

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.
Warning! LUT value is adjusted to account for LUT combining.
Warning! For any ECO changes, please run place_design if there are unplaced

instances

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	_	-	-
0	_	-	Set
0	_	-	Reset
0	_	Set	-
0	_	Reset	-
0	Yes	-	-
6	Yes	-	Set
191	Yes	-	Reset
0	Yes	Set	-
0	Yes	Reset	-

2. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	135	0.00
RAMB36/FIFO*	0	0	0	135	0.00
RAMB18	0	0	0	270	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	0	0	0	240	0.00

4. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	36	0	0	210	17.14
Bonded IPADs	0	0	0	2	0.00
PHY_CONTROL	0	0	0	6	0.00
PHASER_REF	0	0	0	6	0.00
OUT_FIFO	0	0	0	24	0.00
IN_FIFO	0	0	0	24	0.00
IDELAYCTRL	0	0	0	6	0.00
IBUFDS	0	0	0	202	0.00
PHASER_OUT/PHASER_OUT_PHY	0	0	0	24	0.00
PHASER_IN/PHASER_IN_PHY	0	0	0	24	0.00
IDELAYE2/IDELAYE2_FINEDELAY	0	0	0	300	0.00
ILOGIC	0	0	0	210	0.00
OLOGIC	0	0	0	210	0.00

5. Clocking

Site Type	Used	Fixed	Prohibited	Available	Util%
BUFGCTRL	1	0	0	32	3.13
BUFIO	0	0	0	24	0.00
MMCME2_ADV	0	0	0	6	0.00
PLLE2_ADV	0	0	0	6	0.00
BUFMRCE	0	0	0	12	0.00
BUFHCE	0	0	0	96	0.00
BUFR	0	0	0	24	0.00

6. Specific Feature

Site Type	Used	Fixed	Prohibited	Available	Util%
BSCANE2	0	0	0	4	0.00
CAPTUREE2	0	0	0	1	0.00
DNA_PORT	0	0	0	1	0.00
EFUSE_USR	0	0	0	1	0.00
FRAME_ECCE2	0	0	0	1	0.00
ICAPE2	0	0	0	2	0.00
PCIE_2_1	0	0	0	1	0.00
STARTUPE2	0	0	0	1	0.00
XADC	0	0	0	1	0.00

7. Primitives

Ref Name	Used	Functional Category
FDCE	191	Flop & Latch
LUT2	120	LUT
LUT6	54	LUT
CARRY4	46	CarryLogic
LUT5	40	LUT
LUT4	39	LUT
OBUF	30	IO
LUT3	30	LUT
LUT1	6	LUT
IBUF	6	IO
FDPE	6	Flop & Latch
BUFG	1	Clock

8. Black Boxes

Ref Name	Used

9. Instantiated Netlists
